

# Guide: Miniframes Navigation

## 1 Introduction

The `render-miniframes` function generates a navigation bar showing the progress through sections and subsections using “miniframes” (dots or squares).

Previously, it required manual structure extraction. Now, it is designed to be **configured globally** via `navigator-config`, allowing a clean one-line integration in your document.

## 2 Global Configuration

Instead of passing parameters to every `render-miniframes` call, you can set them once at the beginning of your document:

```
#import "@preview/navigator:0.1.6": navigator-config, render-miniframes

#navigator-config.update(c => {
  c.slide-selector = metadata.where(value: (t: "ContentSlide"))
  c.miniframes = (
    fill: navy,
    active-color: white,
    inactive-color: gray,
    style: "compact"
  )
  c
})

// Now you can use a simple one-liner in your header/footer:
#set page(header: context render-miniframes())
```

## 3 Function documentation

`render-miniframes(structure, current-slide-num, ...)`

### 3.1 Parameters Reference

Most parameters default to `auto`, which means they will be resolved from the global `navigator-config` state.

| Option                         | Type                                    | Effect & Expected Values                                                                                            |
|--------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <code>structure</code>         | <code>array</code>   <code>auto</code>  | The presentation structure. If <code>auto</code> , resolved automatically from global <code>slide-selector</code> . |
| <code>current-slide-num</code> | <code>int</code>   <code>auto</code>    | The index of the active slide. If <code>auto</code> , resolved automatically.                                       |
| <code>style</code>             | <code>string</code>   <code>auto</code> | Layout mode: <code>"compact"</code> or <code>"grid"</code> . Defaults to global config.                             |
| <code>marker-shape</code>      | <code>string</code>                     | <code>"circle"</code> (default) or <code>"square"</code> .                                                          |
| <code>marker-size</code>       | <code>length</code>                     | Diameter/width of the markers. Default: 4pt.                                                                        |
| <code>active-color</code>      | <code>color</code>   <code>auto</code>  | Color of the current slide's marker. Defaults to global config.                                                     |
| <code>inactive-color</code>    | <code>color</code>   <code>auto</code>  | Color of future slides' markers. Defaults to global config.                                                         |
| <code>fill</code>              | <code>color</code>   <code>auto</code>  | Background color of the bar block. Defaults to global config.                                                       |
| <code>text-color</code>        | <code>color</code>   <code>auto</code>  | Color of titles. Defaults to contrast with <code>fill</code> .                                                      |
| <code>text-size</code>         | <code>length</code>                     | Size of the titles. Default: 10pt.                                                                                  |

| Option             | Type               | Effect & Expected Values                                                             |
|--------------------|--------------------|--------------------------------------------------------------------------------------|
| font               | string   none      | Font family for titles. Uses document default if none.                               |
| align-mode         | string             | Global horizontal alignment of the block: "left", "center", "right".                 |
| dots-align         | string             | Alignment of the dots <b>within</b> their section column: "left", "center", "right". |
| navigation-pos     | string             | Vertical position of dots relative to titles: "top" or "bottom". Default: "bottom".  |
| show-level1-titles | bool               | Whether to display the names of sections.                                            |
| show-level2-titles | bool               | In grid mode, whether to display subsection names.                                   |
| show-numbering     | bool               | Whether to display heading numbers. Default: false.                                  |
| numbering-format   | string             | Typst numbering format string (e.g., "1.1"). Default: "1.1".                         |
| gap                | length             | Horizontal space between sections. Default: 1.5em.                                   |
| line-spacing       | length             | Vertical space between titles and dots. Default: 4pt.                                |
| inset              | dict   length      | Internal padding of the bar block. Default: (x: 1em, y: 0.5em).                      |
| radius             | length   dict      | Corner rounding of the background block. Default: 0pt.                               |
| width              | length             | Total width of the block. Default: 100%.                                             |
| outset-x           | length             | Horizontal bleed. Useful to make the bar touch page edges.                           |
| max-length         | int   dict   auto  | Maximum length before truncation. Defaults to global config.                         |
| use-short-title    | bool   dict   auto | Whether to use short titles. Defaults to global config.                              |

## 4 Structure Extraction

To work, the navigation bar needs to know the presentation structure. Two functions are provided to extract this data from metadata markers. These are called automatically by `render-miniframes()` if arguments are set to `auto`.

### 4.1 get-structure

`get-structure(slide-selector: auto, filter-selector: none)`

Scans the document for headings and slide markers. Returns a structure dictionary.

### 4.2 get-current-logical-slide-number

`get-current-logical-slide-number(slide-selector: auto, filter-selector: none)`

Determines the index of the current slide relative to the extracted structure.

### 4.3 Selection Parameters

| Option         | Type            | Description                                                                                                           |
|----------------|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| slide-selector | selector   auto | The metadata type used to identify slides. Default is (t: "LogicalSlide"). Useful for custom engines (e.g., Polylux). |

| Option          | Type            | Description                                                                                                                                     |
|-----------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| filter-selector | selector   none | If provided, only pages containing this selector will be counted. Useful to exclude transition slides that might share the same slide metadata. |

## 5 Function Signature

`render-miniframes(structure, current-slide-num, ...)`

### 5.1 The structure object

The structure argument is an array of section dictionaries. Each section has the following schema:

- **Section:** (title: content, loc: location, subsections: array)
- **Subsection:** (title: content, loc: location, slides: array) OR (title: content, loc: location, subsections: array) (if 3 levels are used).
- **Slide:** (number: integer, loc: location)

#### 5.1.1 What is loc?

The loc field expects a Typst **location** object.

- **Purpose:** It defines the destination for navigation links. If a valid location is provided, clicking on the section title or the dot will take the user to that specific position in the PDF.
- **Disabling links:** If set to none, the element will be displayed normally but will not be clickable. This is used in the mock data of this guide.

### 5.2 The current-slide-num argument

An integer representing the current slide number. The function compares this value with the number field of each slide in the structure to determine its state:

- **Active:** `slide.number == current-slide-num`
- **Completed:** `slide.number < current-slide-num`
- **Future:** `slide.number > current-slide-num`

## 6 Basic usage

By default, the navigation bar uses the "grid" style and shows section titles.

```
render-miniframes(structure, 4, use-short-
title: true)
```

| Default Grid Style                         |                                                 |                   |
|--------------------------------------------|-------------------------------------------------|-------------------|
| 1. Intro<br>1.1 Context ● ●<br>1.2 Goals ● | 2. Methods<br>2.1 Raw Data ● ● ●<br>2.2 Tools ● | 3. Results<br>● ● |

## 7 Short Titles & Truncation

Like progressive-outline, Miniframes supports short titles and truncation. This is crucial for navigation bars which have limited horizontal space.

### 7.1 Collecting Short Titles

You must pass the short titles to the structure extractor.

```
#let struct = get-structure(
  all-shorts: query(<short>)
)
```

### 7.2 Truncation & Short Titles

Then, configure the rendering. By default, use-short-title is false.

```
render-miniframes(structure, 4)
```

### 1. Original Titles (Default)

| 1. General                     | 2. Scientific                         | 3. Preliminary |
|--------------------------------|---------------------------------------|----------------|
| Introduction to<br>the Project | Methodology<br>and Data<br>Collection | Results        |
| 1.1 Background                 | 2.1 Raw Data                          |                |
| Context                        | 2.2 Analysis                          |                |
| 1.2 Objectives<br>and Goals    | Tools                                 |                |

```
render-miniframes(
  structure, 4,
  max-length: 12
)
```

### 2. Automatic Truncation

| General Intr... | Scientific M... | Preliminary ... |
|-----------------|-----------------|-----------------|
| 1.1 Backgrou... | 2.1 Raw Data    |                 |
| 1.2 Objectiv... | 2.2 Analysis... |                 |

```
render-miniframes(
  structure, 4,
  use-short-title: true
)
```

### 3. Manual Short Titles

| 1. Intro    | 2. Methods   | 3. Results |
|-------------|--------------|------------|
| 1.1 Context | 2.1 Raw Data |            |
| 1.2 Goals   | 2.2 Tools    |            |

```
render-miniframes(
  structure, 4,
  use-short-title: true,
  max-length: 8
)
```

### 4. Combined: Short Titles + Truncation

| Intro       | Methods     | Results |
|-------------|-------------|---------|
| 1.1 Cont... | 2.1 Raw ... |         |
| 1.2 Goal... | 2.2 Tool... |         |

## 8 Layout Styles

### 8.1 Compact Mode

In "compact" mode, all slide markers of a section are grouped on a single line, regardless of subsections. This is useful for saving space in the header or footer.

```
render-miniframes(
  structure, 4,
  style: 'compact',
  use-short-title: true
)
```

### Compact Mode

| 1. Intro | 2. Methods | 3. Results |
|----------|------------|------------|
|          |            |            |

### 8.2 Grid Mode

The "grid" style is ideal for presentations with many subsections, as it aligns them vertically.

```
render-miniframes(
  structure, 4,
  style: 'grid',
  show-level2-titles: true,
  use-short-title: true
)
```

### Grid Mode with Titles

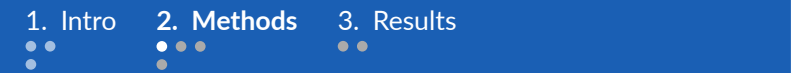
| 1. Intro    | 2. Methods   | 3. Results |
|-------------|--------------|------------|
| 1.1 Context | 2.1 Raw Data |            |
| 1.2 Goals   | 2.2 Tools    |            |

## 8.3 Hiding Titles

You can hide titles at different levels to obtain a minimalist bar.

```
render-miniframes(  
  structure, 4,  
  style: 'grid',  
  show-level2-titles: false,  
  use-short-title: true  
)
```

### Hiding Subsection Titles (Grid)



1. Intro    2. Methods    3. Results

The navigation bar is blue with white text. It has three main sections: '1. Intro', '2. Methods', and '3. Results'. Each section has a progress indicator below it: '1. Intro' has two dots, '2. Methods' has three dots, and '3. Results' has two dots.

```
render-miniframes(  
  structure, 4,  
  show-level1-titles: false,  
  use-short-title: true  
)
```

### Hiding Section Titles (Dots Only)



1.1 Context    2.1 Raw Data    1.2 Goals    2.2 Tools

The navigation bar is blue with white text. It has four main sections: '1.1 Context', '2.1 Raw Data', '1.2 Goals', and '2.2 Tools'. Each section has a progress indicator below it: '1.1 Context' has two dots, '2.1 Raw Data' has three dots, '1.2 Goals' has one dot, and '2.2 Tools' has two dots.

## 9 Customization

### 9.1 Markers

Change the shape and size of the progress indicators.

```
render-miniframes(  
  structure, 4,  
  marker-shape: 'square',  
  marker-size: 6pt,  
  use-short-title: true  
)
```

### Square Markers



1. Intro    2. Methods    3. Results

1.1 Context    2.1 Raw Data    1.2 Goals    2.2 Tools

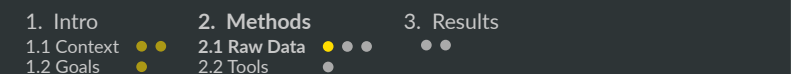
The navigation bar is blue with white text. It has four main sections: '1. Intro', '2. Methods', '3. Results', and a fourth section. Each section has a progress indicator below it: '1. Intro' has two squares, '2. Methods' has three squares, '3. Results' has two squares, and the fourth section has one square.

### 9.2 Colors & Typography

Fine-tune the appearance of markers and labels.

```
render-miniframes(  
  structure, 4,  
  active-color: yellow,  
  inactive-color: gray,  
  text-color: luma(200),  
  text-size: 8pt,  
  fill: rgb('#2d3436'),  
  use-short-title: true  
)
```

### Colors & Fonts



1. Intro    2. Methods    3. Results

1.1 Context    2.1 Raw Data    1.2 Goals    2.2 Tools

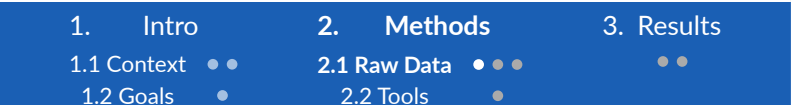
The navigation bar is dark gray with yellow text. It has four main sections: '1. Intro', '2. Methods', '3. Results', and a fourth section. Each section has a progress indicator below it: '1. Intro' has two yellow dots, '2. Methods' has three yellow dots, '3. Results' has two yellow dots, and the fourth section has one yellow dot.

### 9.3 Alignment & Spacing

Control the rhythm and positioning of the navigation elements.

```
render-miniframes(  
  structure, 4,  
  align-mode: 'center',  
  dots-align: 'center',  
  gap: 3em,  
  line-spacing: 8pt,  
  use-short-title: true  
)
```

### Centered & Airy



1. Intro    2. Methods    3. Results

1.1 Context    2.1 Raw Data    1.2 Goals    2.2 Tools

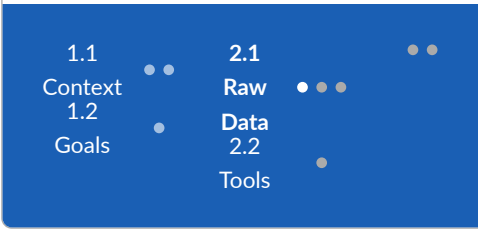
The navigation bar is blue with white text. It has four main sections: '1. Intro', '2. Methods', '3. Results', and a fourth section. Each section has a progress indicator below it: '1. Intro' has two dots, '2. Methods' has three dots, '3. Results' has two dots, and the fourth section has one dot.

### 9.4 Advanced Layout

Use inset and width to integrate the bar into specific layout zones.

```
render-miniframes(  
  structure, 4,  
  width: 60%,  
  align-mode: 'center',  
  inset: 15pt,  
  show-level1-titles: false,  
  use-short-title: true  
)
```

Compact Centered Bar



```
render-miniframes(  
  structure, 4,  
  radius: 10pt,  
  fill: rgb('#34495e'),  
  inset: (x: 2em, y: 1em),  
  use-short-title: true  
)
```

Rounded Corners

